Personal History Statement

I was about eight when I asked my grandpa what his favorite molecule was. He was a PhD chemist, so I expected some complicated compound I'd struggle to pronounce. Instead, he simply said, "Oh, that's easy, water!" At the time, this answer confused me. Water was everywhere. It was just a standard part of life, mundane even. "You'll be surprised how even the most simple things become the most beautiful once you truly understand them," he explained. Curious to test his statement, I took apart my dad's car radio, hoping to figure it out and see its beauty. But, instead, I was left with a broken radio and an upset father. Nonetheless, I continued my search, delving into books on famous inventors and scientists. At the time, I was most drawn to Isaac Newton. Not just because we shared a name, but mainly because I was fascinated with his discoveries in optics. Learning that white light was composed of a continuous spectrum of light astonished me; lightbulbs in my classroom and the sunlight reflecting off the playground were all enclosing a secret rainbow, a hidden world right in front of me.

Finally, in high school, I got the chance to study physics formally. I was amazed at how one could accurately predict the trajectories of objects using simple calculations. With this newfound knowledge, I attempted to construct a rocket and predict its motion, though I nearly set my garage on fire while testing my potassium chlorate fuel mixture. At this point, I was certain I wanted to study and research physics at a high level, and I was fortunate enough to be accepted into a research university like Northwestern and receive the scholarships I needed to attend.

Once there, I was invited to apply to the Integrated Science Program (ISP), a selective program that aims to develop a highly interdisciplinary community and understanding of the sciences, allowing students to study a mix of STEM courses from organic chemistry to quantum mechanics, while creating an accelerated path towards one's particular interests. However, it was not all sunshine and rainbows. As a first-generation college student, I didn't know what to expect or how to keep up with the rigor and the fast pace of Northwestern, and especially ISP. And to make it even worse, I lost my biggest inspiration, my grandfather, in the middle of my freshman year–the person who first inspired me to get into science. This was a really hard time for me, and I struggled to focus in my classes.

I remember at his funeral hearing all the amazing stories about him, how kind and humble he was, always willing to talk about his research to those who wanted to know, but always being able to simplify it to meet anyone where they were. Hearing from his younger colleagues that he mentored and about the influence he had on them really inspired me. I always knew my grandpa was a great inspiration to me, but hearing the impact he had on so many others made me want to become just like him even more than before. When I got back to campus, I was met with support from my friends in ISP, who helped me catch up on what I missed in class and get back on track. And now, I am even more motivated to continue my studies and reach a point where I can have the same impact as my grandpa, both as a researcher and as a mentor.

The Integrated Science Program has been one of the most valuable peer groups I have been part of during my time at Northwestern. Many of the friends I have developed in the community I have gone to for advice in my lab, such as asking an ISP electrical engineer for help with a filter and amplifier for data acquisition, and an ISP mechanical engineer to teach me SolidWorks. Likewise, I have helped my ISP electrical engineering friend learn about optical setups. Even just going to weekly trivia with the friends I have made in the program and discussing the vastly different types of research we are working on has helped me develop a broader picture and generate new ideas. This program has taught me how vital a diverse and collaborative environment is for science to thrive. Even though I won't graduate with the ISP degree due to a few final class conflicts, it has given me a strong network and group of friends that I know I will always be able to rely on throughout my future career as a researcher.

In Professor Odom's lab, I have played an important role in mentoring two other undergraduates. I have spent much time working with and talking to them to develop an understanding of our lab's goals and how their skills and interests can best be used to achieve them. I have also spent much time talking to graduate students to help figure out how new students' skills can best be utilized and what possible projects they could take on. I have also used my background and love of the arts to help create scientific illustrations that my lab now has on our home and research pages to better communicate our project and science to others. I have deeply enjoyed the chance to mentor, as I remember how formative the mentorship I received from my first work with a graduate student was, and I am excited for the opportunity to continue and improve at mentoring in graduate school.

Throughout my time at Northwestern, I have learned so much through classes and research, and I feel I am getting closer to uncovering the true beauty my grandfather talked about. I'm sad I will never be able to speak with him about the beauty I see, and discuss if it's any different from his version. But I am hopeful to one day share what I see with others and maybe spark their desire for science and a quest to discover their own beauty in the world.